

Akshit Gajera

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Professional Summary

Data Science undergraduate with a strong foundation across the full machine learning lifecycle, including data analysis, statistical modeling, deep learning, and Generative AI system design. Skilled in building end-to-end ML pipelines, LLM-powered RAG applications, and interactive dashboards, with hands-on experience applying MLOps practices such as containerization, CI/CD automation, and model monitoring for reliable, production-ready deployment. Proficient in Python, SQL, scikit-learn, TensorFlow, LangChain, Docker, and Power BI.

Technical Skills

Languages: Python, SQL

Data Science & ML: NumPy, pandas, scikit-learn, Feature Engineering, EDA, Statistical Modeling, Model Evaluation

Deep Learning: TensorFlow, Keras, ANN, CNN, RNN, LSTM, GRU, Encoder-Decoder, Transformers, Attention Mechanisms

Generative AI & NLP: LLMs, RAG Pipelines, LangChain, LangGraph, Prompt Engineering, NLP, NLTK, Vector Databases, Semantic Search

MLOps: Docker, Kubernetes, CI/CD Pipelines, GitHub Actions, MLflow, Evidently AI, Prometheus, Grafana, Model Monitoring & Deployment

Visualization & BI: Matplotlib, Seaborn, Power BI, Excel

Deployment: Flask, FastAPI, Streamlit

Databases: MySQL, PostgreSQL

Tools: Git, GitHub, Jupyter Lab, Google Colab, VS Code

Projects

AI-Powered Video Intelligence & Semantic Search System

[GitHub](#) | [Live Demo](#)

- Designed a fully localized 4-phase RAG pipeline converting raw MP4 course videos into a queryable AI knowledge base, eliminating manual video scrubbing for students.
- Used FFmpeg for audio extraction and OpenAI Whisper (large-v2) for transcription, storing structured JSON with precise start/end timestamps for each segment.
- Generated semantic embeddings with Ollama bge-m3 and built a cosine similarity engine to surface Top-5 relevant chunks, injected into llama3.2 for timestamp-accurate answers via Streamlit.

LLM-Based Automated Code Review & PR Intelligence System

[GitHub](#)

- Built an autonomous AI agent triggered on every GitHub pull request via GitHub Actions CI/CD workflows, analyzing code diffs and posting structured review comments without human intervention.
- Engineered a multi-category prompt layer feeding PR diff context into Gemini API, parsing structured output across 4 dimensions: critical bugs, security vulnerabilities, code style, and performance.
- Implemented a Gmail SMTP notification pipeline delivering formatted review reports, including total issues, severity breakdown, and cleanup recommendations, to repository maintainers.

IMDB Sentiment Analysis Engine — RNN/LSTM/GRU Comparison

[GitHub](#) | [Live Demo](#)

- Built and benchmarked Simple RNN, LSTM, and GRU architectures for binary sentiment classification on 50,000 IMDB movie reviews, selecting GRU for the best convergence speed and generalization.
- Engineered an NLP pipeline (HTML/URL/punctuation removal, tokenization, stopword removal, lemmatization) with a Keras tokenizer using a 10,000-word vocabulary and 300-token sequence padding.
- Achieved 88% accuracy, 90% precision, and 87% recall on ~10,000 held-out reviews; deployed via Streamlit with real-time prediction, confidence scoring, and batch CSV/Excel processing.

Education

Gujarat Technological University (GTU) — B.Tech in Computer Science 2023 – 2027

SSASIT, Surat, Gujarat | GPA: 9.0 / 10.0

Certification

Data Science and Analytics with GenAI — Sheryians Coding School Jan 2026